

INFO8006 Introduction to Artificial Intelligence

Exam of August 2026

Instructions

- The exam lasts for 4 hours.
- You are allowed to use a calculator during the exam, but documents of any kind are forbidden.
- The last two pages can be used for scratch work or for extra space. If you want work done there to be graded, mention where to look **in big letters with a box around them**, on the page with the question.
- Write your last name, first name, and ULiège ID on the first page. Write only your ULiège ID on all the other pages.
- Before handing in your exam, **sort all the pages according to the page numbers** (even if you used additional pages to answer a question).

Good luck!

LAST NAME, FIRST NAME (in uppercase): _____

ULiège ID (s201234): _____

Question 1 [4 points] Multiple choice questions. Choose one of the four choices by filling in its circle. Correct answers are graded $+\frac{4}{10}$, wrong answers are graded $-\frac{2}{15}$ and the absence of answers is graded 0. The total of your grade for Question 1 is bounded below at 0/4.

1. The 2048 game is an example of a ...
 - ☐ fully observable, deterministic game.
 - ☐ partially observable, deterministic game.
 - ☐ fully observable, stochastic game.
 - ☐ partially observable, stochastic game.
2. Let h_1 and h_2 be two admissible heuristics. Which of the following is true?
 - ☐ $\max(h_1, h_2)$ is admissible and dominates both h_1 and h_2 .
 - ☐ $h_1 + h_2$ is admissible.
 - ☐ $\min(h_1, h_2)$ dominates both h_1 and h_2 .
 - ☐ $h_1 \times h_2$ is admissible.
3. Let X_1 , X_2 and X_3 be the results of three independent six-sided dice rolls. Let $S_2 = X_1 + X_2$ and $S_3 = X_1 + X_2 + X_3$. Which of the following is true?
 - ☐ $X_1 \perp S_2$
 - ☐ $X_1 \perp S_3 \mid S_2$
 - ☐ $X_1 \perp S_3$
 - ☐ $S_2 \perp S_3 \mid X_1$
4. In a hidden Markov model, assume that the sensor is uninformative, i.e. $P(e|x) = P(e)$ for all states x . Then the belief state $\mathbf{P}(X_t|e_{1:t})$...
 - ☐ is uniform over the states.
 - ☐ is equal to the prior $\mathbf{P}(X_0)$.
 - ☐ is equal to the prediction obtained by pushing $\mathbf{P}(X_{t-1}|e_{1:t-1})$ through the transition model.
 - ☐ cannot be computed.
5. Fitting a regression model $P(y|\mathbf{x}) = \mathcal{N}(y; \mu = \mathbf{w}^T \mathbf{x} + b, \sigma^2)$ by maximum likelihood is equivalent to ...
 - ☐ minimizing the zero-one loss.
 - ☐ minimizing the cross-entropy loss.
 - ☐ minimizing the likelihood of the parameters.
 - ☐ minimizing the mean squared error.
6. In machine learning, the loss is evaluated on a separate test set because ...
 - ☐ the test loss is always smaller than the training loss.
 - ☐ a small training loss does not imply that the model generalizes to new data.
 - ☐ gradient descent cannot be run on the whole training set.
 - ☐ the training loss is not differentiable.

7. Let $\mathbf{x} = (2, 0, 0, 4, 0, 0, 2)$ and $\mathbf{u} = (1, -1)$. The convolution $\mathbf{x} \circledast \mathbf{u}$ (as usually defined in convolutional networks) is equal to ...
- ☐ $(2, 0, 0, 4, 0, 0, 2)$
 - ☐ $(-2, 0, 4, -4, 0, 2)$
 - ☐ $(2, 0, -4, 4, 0, -2)$
 - ☐ $(0, 0, 0, 0, 0, 0)$
8. In a neural network trained for classification over $C > 2$ classes, the output layer is usually ...
- ☐ a single unit with a sigmoid activation.
 - ☐ C units with a softmax activation.
 - ☐ C units with a ReLU activation.
 - ☐ C units with no activation function.
9. The Bellman optimality equations generalize to state-action-value $Q(s, a)$ as ...
- ☐ $Q(s, a) = R(s) + \gamma \sum_{s'} P(s'|s, a) Q(s', a)$.
 - ☐ $Q(s, a) = R(s) + \gamma \sum_{s'} P(s'|s, a) Q(s', \pi(s'))$.
 - ☐ $Q(s, a) = R(s) + \gamma \sum_{s'} P(s'|s, a) \max_{s'} Q(s', a')$.
 - ☐ $Q(s, a) = R(s) + \gamma \sum_{s'} P(s'|s, a) \max_{a'} Q(s', a')$.
10. Temporal-difference learning differs from Monte Carlo evaluation in that ...
- ☐ it updates the value function after each step.
 - ☐ it requires a model of the environment.
 - ☐ it always converges faster.
 - ☐ it does not require exploration mechanisms.

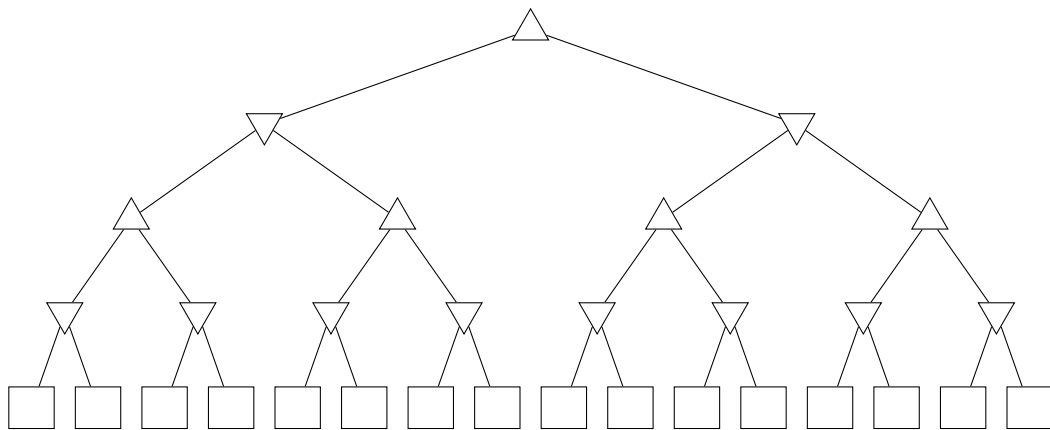
Question 2 [4 points] Five coins are laid in a row on a table:

3 5 6 1 12

Two players, MAX and MIN, play the following game. Starting with MAX, they alternately remove either the leftmost or the rightmost of the remaining coins. After four removals a single coin is left: the utility of MAX is the value of that coin, and the utility of MIN is its opposite.

(a) Formalize this game as a search problem.

(b) The complete game tree of this game is given below, where triangles pointing up are MAX nodes, triangles pointing down are MIN nodes, and squares are terminal states. At every node, the left branch removes the leftmost remaining coin and the right branch removes the rightmost one.



Fill in the utility of each terminal state, then compute the minimax value of every node of the tree. Which coin should MAX remove first?

- (c) MIN is in fact a beginner who removes one of the two end coins with equal probability. Which algorithm should MAX use instead? Compute the value of every node again and give the coin that MAX should now remove first.

- (d) MIN is now a teammate who maximizes the same utility as MAX. Which coin should MAX remove first, and what utility does MAX obtain?

- (e) MAX is now wrong about MIN, but only for the very first removal. Consider the two following cases:

- (i) MAX believes that MIN is a beginner, and therefore removes the coin found in (c). In reality, MIN plays optimally.
- (ii) MAX believes that MIN plays optimally, and therefore removes the coin found in (b). In reality, MIN is a beginner.

In both cases, MAX discovers the mistake right after this first removal, and plays the rest of the game with the correct algorithm. What utility does MAX obtain in each case? What do you conclude?

Question 3 [4 points] A meteorological station monitors weather conditions. The true temperature T (in °C) at its location is modeled as a Gaussian random variable with prior distribution $T \sim \mathcal{N}(20, 16)$ (mean 20°C, variance 16).

Two sensors provide noisy measurements of the temperature:

- Sensor 1 gives reading $S_1|T \sim \mathcal{N}(T, 4)$,
- Sensor 2 gives reading $S_2|T \sim \mathcal{N}(T, 1)$.

Additionally, humidity H (in %) depends on temperature according to $H|T \sim \mathcal{N}(3T + 10, 25)$.

- (a) Draw the Bayesian network corresponding to this model. Specify all nodes and their associated probability distributions.

- (b) Answer the following independence questions. Justify your answers using d-separation.

- (i) Are S_1 and S_2 marginally independent? ($S_1 \perp S_2$?)
- (ii) Are S_1 and S_2 conditionally independent given T ? ($S_1 \perp S_2|T$?)
- (iii) Are S_1 and H conditionally independent given S_2 ? ($S_1 \perp H|S_2$?)

(c) Using the Gaussian identities in Appendix A, derive the posterior distribution $p(T = t|S_1 = 26)$.

(d) Now suppose both sensors give readings: $S_1 = 26$ and $S_2 = 22$. Derive the posterior distribution $p(T = t|S_1 = 26, S_2 = 22)$.

(e) Derive the predictive distribution $p(H = h|S_1 = 26)$.

Question 4 [4 points] In a greenhouse, a fungal infection can spread through an entire crop. Every morning, a grower runs a rapid strip test on a plant in order to decide whether or not to spray it with a fungicide. However, the test is noisy: it returns a positive result in 90% of cases if the plant is infected, and a negative result in 70% of cases if the plant is healthy.

We model the state of a plant at time t as $X_t \in \{\text{healthy}, \text{infected}\}$ and the test result as evidence $E_t \in \{+, -\}$. At $t = 0$, the plant is still at the nursery, with uniform probability of being healthy or infected. It is then delivered to the greenhouse, where it spends a day in its crate before being inspected. During this time, no test is performed and no fungicide is sprayed. The dynamics without fungicide are as follows: an infected plant remains infected with certainty, while a healthy plant has a 20% chance of becoming infected. Monitoring begins at $t = 1$.

- (a) Define the prior distribution $\mathbf{P}(X_0)$ and the transition model $\mathbf{P}(X_1|X_0)$. Derive the predicted distribution $\mathbf{P}(X_1)$ before any observation.

- (b) Define the observation model $\mathbf{P}(E_t|X_t)$. The first strip test at $t = 1$ returns negative. Derive $\mathbf{P}(X_1|e_1 = -)$.

- (c) From $t = 1$ onwards, the grower decides on an action $a_t \in \{\text{spray}, \text{no spray}\}$ after observing the test result e_t . The action affects the transition from X_t to X_{t+1} :
 - If the fungicide is sprayed ($a_t = \text{spray}$): an infected plant has a 60% chance of becoming healthy, while a healthy plant has a 10% chance of becoming infected (the fungicide weakens its natural defences).
 - If the fungicide is not sprayed ($a_t = \text{no spray}$): dynamics are as described above.
 Draw the (dynamic) Bayesian network representing the model over the time steps $t = 0, 1, 2, 3$, including the state, evidence and action nodes.

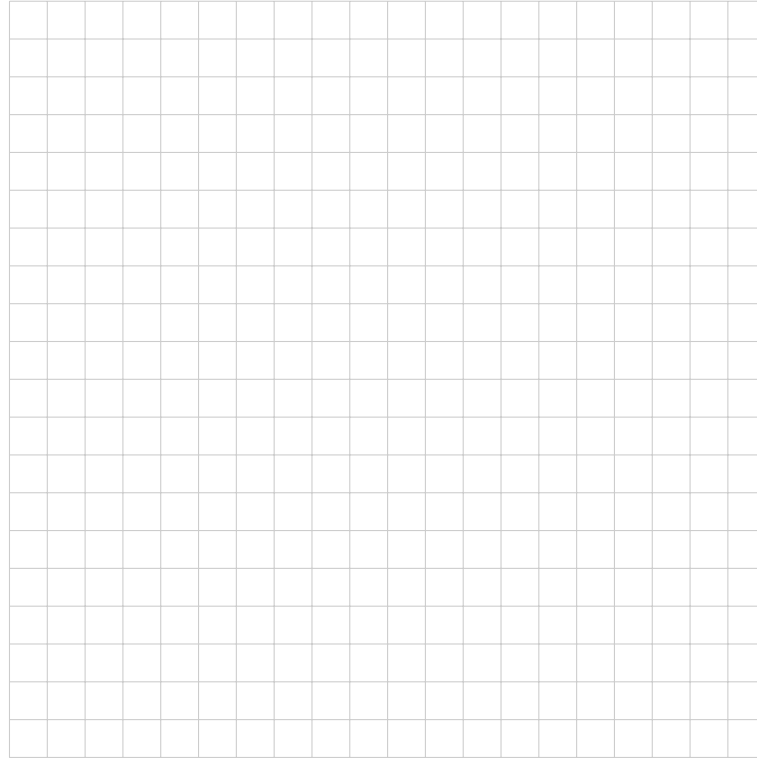
- (d) Define the transition model $\mathbf{P}(X_{t+1}|X_t, a_t)$. Derive the general filtering recursion for $\mathbf{P}(X_t|e_{1:t}, a_{1:t-1})$ in terms of the model components.

- (e) Starting from $\mathbf{P}(X_1|e_1)$ computed in part (b), compute the belief states $\mathbf{P}(X_t|e_{1:t}, a_{1:t-1})$ for $t = 2, 3$, given the evidence-action pairs

$$(e_1, a_1) = (-, \text{no spray}), \quad (e_2, a_2) = (+, \text{spray}), \quad (e_3, a_3) = (-, \text{spray}).$$

- (f) After some thought, the grower wonders whether the plant might already have been infected when the test returned negative at $t = 1$, and wants to compute the smoothing distribution $\mathbf{P}(X_1|e_{1:3}, a_{1:2})$. Discuss whether $a_1 = \text{no spray}$ was a good decision in hindsight, based on the smoothing result.

- (c) Plot the decision boundary $\hat{y} = 0.5$ of the classifier on the grid below. Clearly indicate which region(s) are classified as class 1 and which as class 0.



- (d) We now consider a modified network with a third hidden unit:

$$h_1 = \text{ReLU}(x_1 - 1)$$

$$h_2 = \text{ReLU}(x_2 - 2)$$

$$h_3 = \text{ReLU}(x_1 + x_2 - 8)$$

$$\hat{y} = \sigma(h_1 + h_2 - h_3 - 2)$$

On the same grid, plot the new decision boundary as a dashed line and compare it to the original one. What is the effect of adding h_3 ?

Appendix A. Gaussian identities (Särkkä, 2013).

(a) If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}\right) = \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^T & \mathbf{B} \end{pmatrix}\right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\mathbf{a}, \mathbf{A}) \\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y}|\mathbf{b}, \mathbf{B}) \\ p(\mathbf{x}|\mathbf{y}) &= \mathcal{N}(\mathbf{x}|\mathbf{a} + \mathbf{CB}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{CB}^{-1}\mathbf{C}^T) \\ p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}(\mathbf{y}|\mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}). \end{aligned}$$

(b) If the random variables $\mathbf{x}$ and $\mathbf{y}$ have Gaussian probability distributions

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\mathbf{m}, \mathbf{P}) \\ p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}(\mathbf{y}|\mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R}), \end{aligned}$$

then the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ is Gaussian with

$$p\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}\right) = \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{m} \\ \mathbf{H}\mathbf{m} + \mathbf{u} \end{pmatrix}, \begin{pmatrix} \mathbf{P} & \mathbf{P}\mathbf{H}^T \\ \mathbf{H}\mathbf{P} & \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R} \end{pmatrix}\right).$$

Extra page 1 / 2.

Extra page 2 / 2.